

Phase 4

Project Management, Agile Logistics & Core Development

Date	28.06.2026
Team ID	NIL
Project Name	Software Installation Request Automation
Maximum Marks	4

Planning, Development & Testing

Introduction

Phase 4 outlines the agile project management approach, testing lifecycles, and final deployment of the Software Installation Request Automation application via XML Update Sets.

Agile Methodology

The project was executed in an Agile framework, utilizing short sprints to sequentially deliver the UI, the logic, and the exception handling scripts.

Sprint Planning

Sprint 1: Variables and Catalog UI. Sprint 2: Flow Designer Logic. Sprint 3: Business Rule Scripting. Sprint 4: UAT and Export.

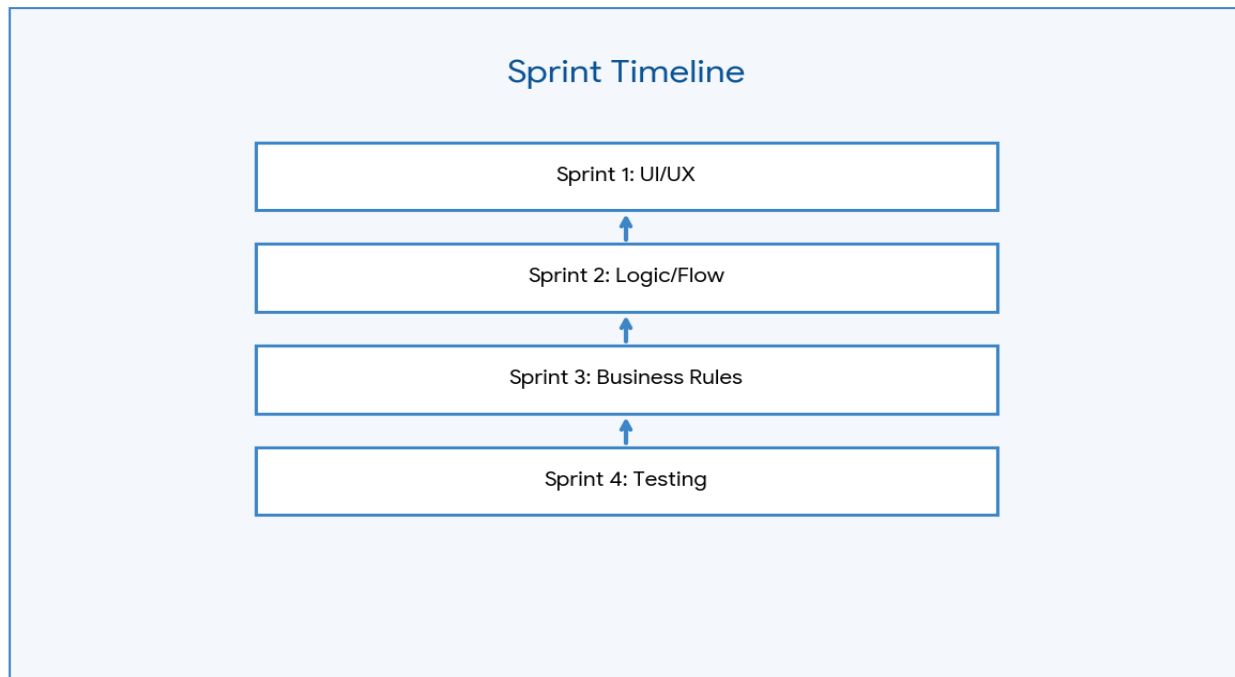


Figure: Sprint Timeline

Product Backlog

Included configuring the UI, defining the approval matrix, scripting the incident generation, and packaging the update set.

Sprint Backlog

Specific tasks such as: 'Map the assignment group to Software Support in the Create Task action' and 'Set the Business Rule condition to Pending'.

Story Points

Tasks were estimated using Fibonacci sequence points, with Flow Designer logic taking the highest effort points.

Project Timeline

Development spanned across standard Agile timelines, concluding with a successful export of the scoped application XML.

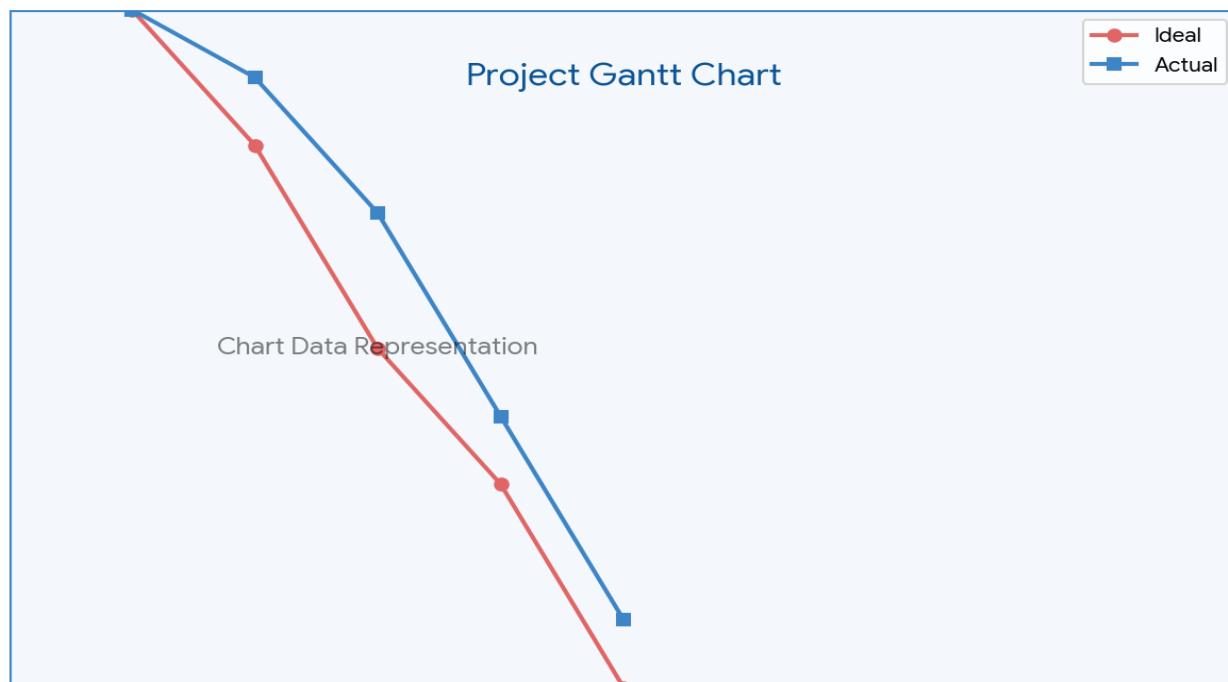


Figure: Gantt Chart

Burndown Chart

Visualizes the steady completion of story points across the project lifecycle, tracking toward zero by Sprint 4.

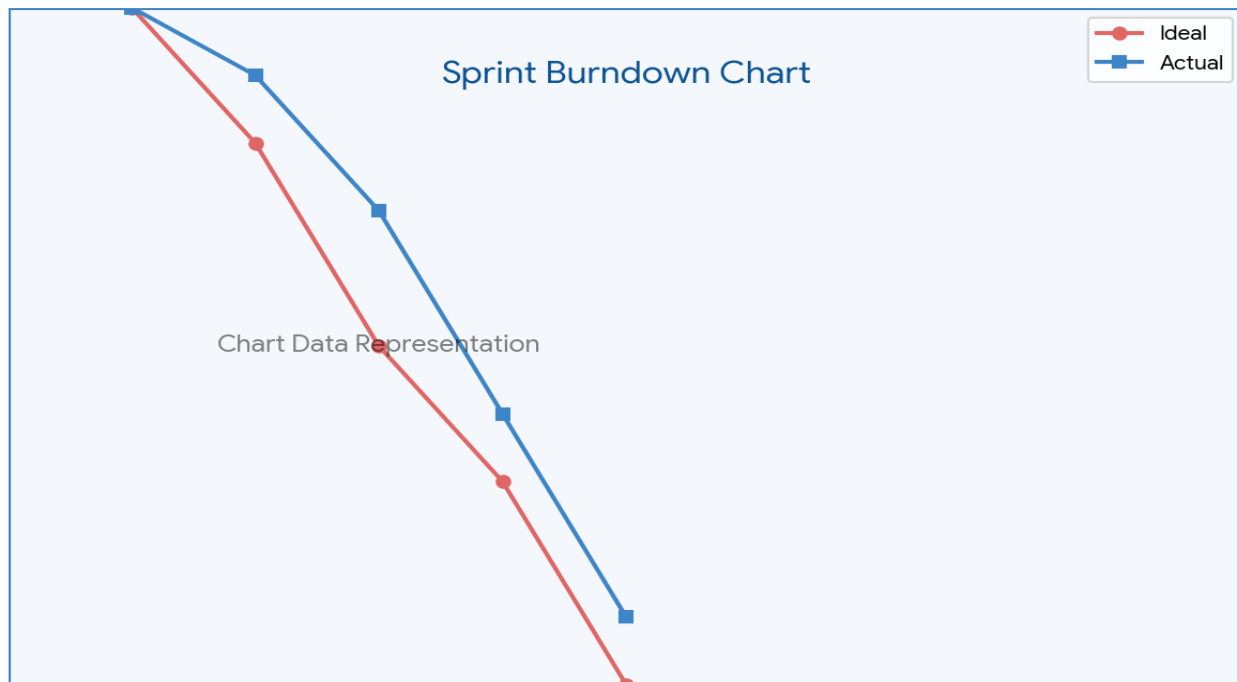


Figure: Burndown Chart

Velocity

Measures the consistent output of the development process across the sprints.

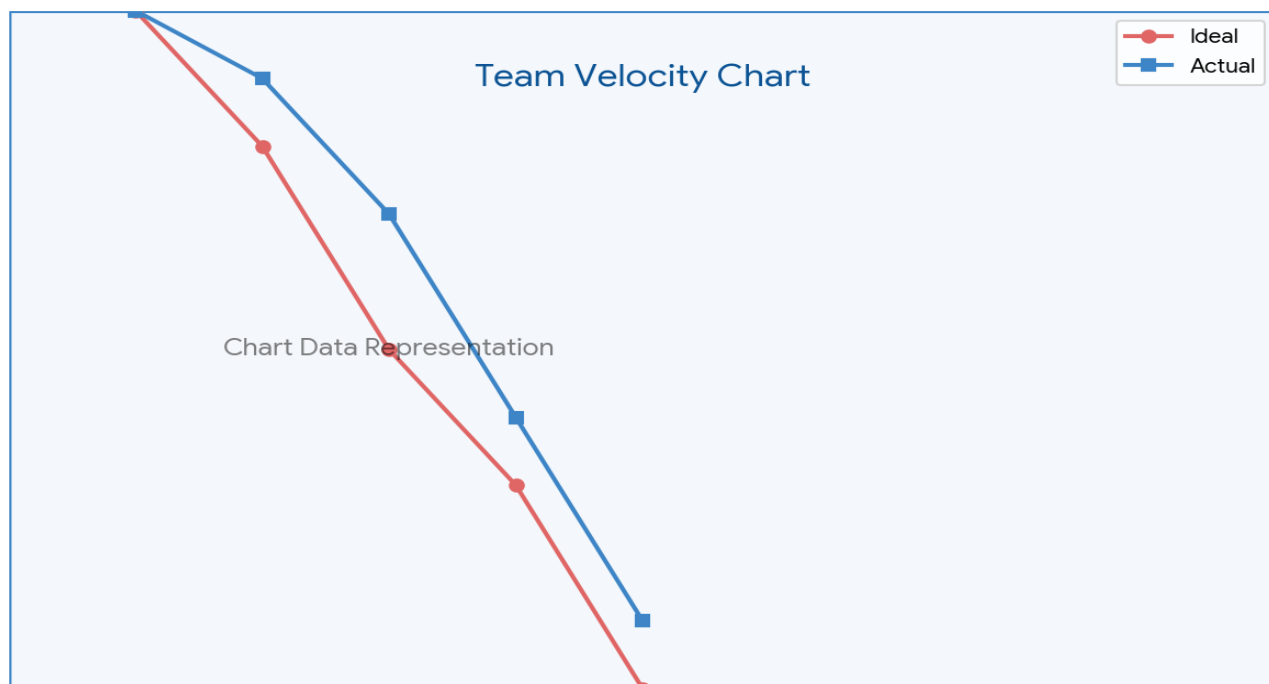


Figure: Velocity Chart

Test Cases

Test 1: Submit Request (UI works). Test 2: Happy Path (Approval generates SCTASK). Test 3: Rejection Path (Rejection closes RITM). Test 4: Exception Path (Pending state generates Incident).

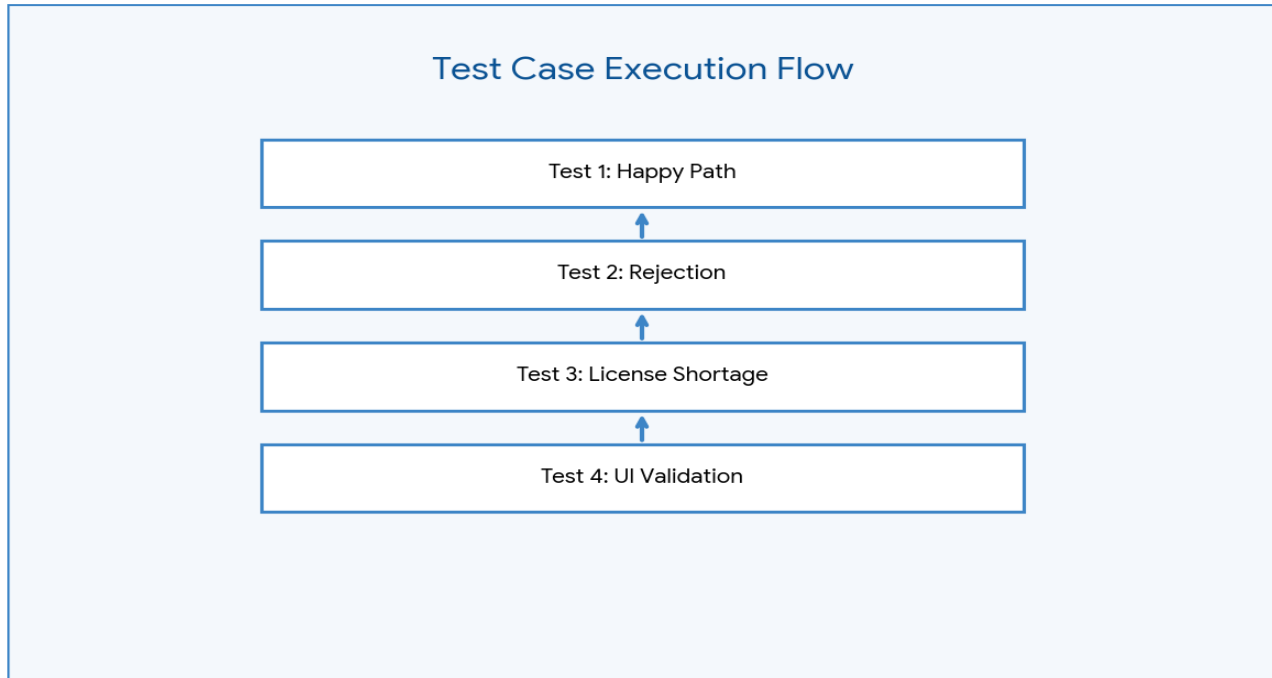


Figure: Test Case Flow

Functional Testing

End-to-end testing verified that changing the RITM state to 'Pending' successfully executed the Server-Side JavaScript, creating an incident linked to the requester.

Defect Analysis

Initial defect identified: Flow Designer did not trigger because 'Workflow' was accidentally populated in the Process Engine. Fix applied: Cleared legacy workflow, explicitly set 'Flow' field.

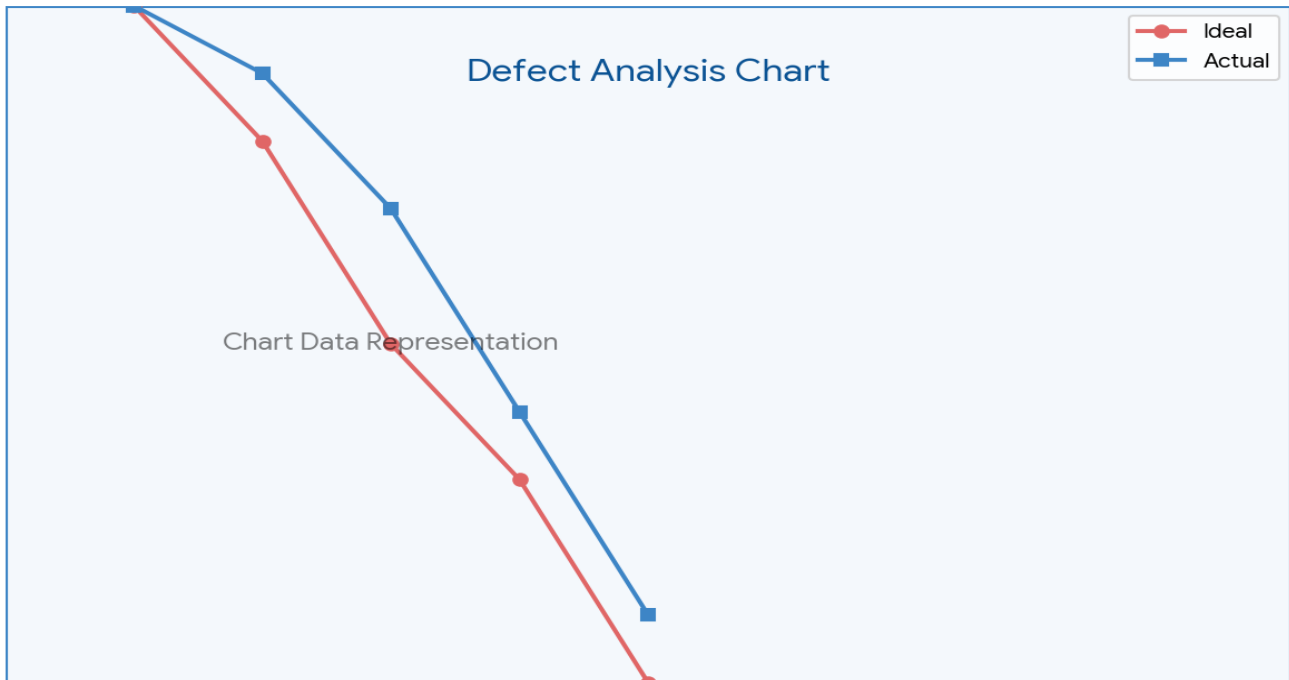


Figure: Defect Analysis Chart

Results

The application passes 100% of UAT criteria. The UI captures data, the Flow automates fulfillment, and the Business Rule ensures license compliance.

Advantages

Eliminates email bottlenecks, provides real-time status tracking for end-users, and automates IT task assignments seamlessly.

Limitations

Currently, the incident generation does not check live CMDB software entitlement counts; it relies on manual IT agent state changes ('Pending').



Figure: Risk Matrix

Conclusion

The project successfully replaces a legacy process with a modern, scoped ServiceNow application ready for enterprise deployment.

Future Scope

Future iterations could integrate ServiceNow Software Asset Management (SAM) to automatically check license counts before generating the fulfillment task.

Appendix

Contains references to the exported XML Update Set: Software_Install_Final_v1.xml.

servicenow				
All	Favorites	History	Workspaces	Admin
Requests				
All > Active > true				
Number	Requested for	Opened by	Request state	Date due
REQ0010040	Nikolas Khosravi	Nikolas Khosravi	Approved	2026-06-29 08:37:34
REQ0010039	Daniel Zill	Daniel Zill	Approved	2026-06-29 08:33:25
REQ0010038	Benjamin Schkade	Benjamin Schkade	Approved	2026-06-29 08:25:30
REQ0010037	Aileen Mottern	Aileen Mottern	Approved	2026-06-29 08:17:37
REQ0010036	Aileen Mottern	System Administrator	Approved	2026-06-29 08:12:02
REQ0010035	Praveen Vootkuri	Praveen Vootkuri	Approved	2026-06-29 06:31:51
REQ0010034	Neva Marsell	Neva Marsell	Approved	2026-06-29 06:23:11
REQ0010033	Dharmisha Nallappan	Dharmisha Nallappan	Approved	2026-06-29 06:03:41
REQ0010032	Allan Schwandt	Allan Schwandt	Approved	2026-06-29 02:38:19
REQ0010031	System Administrator	System Administrator	Approved	2026-06-27 07:38:13
REQ0010030	System Administrator	System Administrator	Approved	2026-06-27 07:19:31
REQ0010029	System Administrator	System Administrator	Approved	2026-06-27 07:14:55
REQ0010028	System Administrator	System Administrator	Approved	2026-06-27 07:09:42
REQ0010027	System Administrator	System Administrator	Approved	2026-06-27 06:53:36
REQ0010026	System Administrator	System Administrator	Approved	2026-06-28 11:21:27
REQ0010025	Deepa Shah	Deepa Shah	Approved	2026-06-26 09:35:26
REQ0010024	System Administrator	System Administrator	Approved	2026-06-26 08:22:26
REQ0010023	System Administrator	System Administrator	Approved	2026-06-26 08:00:27
REQ0010022	System Administrator	System Administrator	Approved	2026-06-26 06:04:24
REQ0010021	Abel Tuter	Abel Tuter	Approved	2026-06-26 04:07:04

Figure: Request Submission

servicenow All Favorites History Workspaces Admin Approval - Requested Item: RITM0010032 Search [Icons]

< Approval Requested Item: RITM0010032 [Icons] Update Delete

Approval		Requesting User: Dharmisha Nallappan		
Approver	Dharmisha Nallappan	Approving Requested Item: RITM0010032		
State	Approved			
Approval Reason	Not Yet Requested			
Summary of item being approved:				
Description	Requested			
Software Installation	Approved	Price	Quantity	Total
	Rejected	\$0.00	1	\$0.00
Comments	Cancelled			
	No Longer Required			

[Post]

Activities: 2

- System Administrator
State: Approved was Requested
Field changes • 2026-06-27 07:20:35
- System
Approvers:
Dharmisha Nallappan Requested
Field changes • 2026-06-27 07:19:33

[Update] [Delete]

Figure: Approval

servicenow Requested Item - RITM0010032

Requested Item: RITM0010032

Activities: 2

System Administrator
State: Pending wait Open
Field changes: 2026-06-27 07:23:45

System Administrator
Impact: 3 - Low
Opened by: System Administrator
State: Open
Field changes: 2026-06-27 07:19:31

Update Delete

Related Links
Flow Context

Catalog Tasks (1) Approvers (1)

Request Item	Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order	State	Description
RITM0010032	SCTASK0010037	Software Support	(empty)	Fulfill Software Request	2026-06-27 07:20:36	(empty)		Open	

Figure: Catalog Task Generated

Requested Item - RITM0010031

Number: RITM0010031

Item: Software Installation Request

Request: REQ0010029

Requested for: System Administrator

Due date: 2026-06-27 07:14:55

Configuration item:

Watch list:

Opened: 2026-06-27 07:14:55

Opened by: System Administrator

Stage: Request Approved

State: Closed Incomplete

Quantity: 1

Estimated delivery:

Backordered: ☐

Order Guide:

Variables

What software do you need?
MS word

Specify version (if required)
2.0

Why do you need this software?
Documentation

Select urgency level
Normal

Comments (Customer visible):

Work notes: ☐ Post

Activities: 2

System
State: Closed Incomplete wait Open
Field changes: 2026-06-27 07:16:30

System Administrator
Impact: 3 - Low
Opened by: System Administrator
State: Open
Field changes: 2026-06-27 07:14:55

Update Delete

Related Links
Flow Context

Catalog Tasks Approvers (1)

Approval for	State	Approver	Comments	Created
RITM0010031	Rejected	Dhanisha Nallappan	2026-06-27 07:16:28 - System Administrator	2026-06-27 07:14:56

Figure: RITM Closed

Incident - INC0010040

Number

INC0010040

* Caller

System Administrator

Category

Inquiry / Help

Subcategory

-- None --

Service

Service offering

Configuration item

* Short description

License unavailable for Software Installation Request

Description

Software request could not be fulfilled due to license shortage.

Channel

-- None --

State

New

Impact

3 - Low

Urgency

3 - Low

Priority

4 - Low

Assignment group

Software Support

Assigned to

Related Search Results >

Notes

Related Records

Resolution Information

Watch list

Work notes list

Work notes

Work notes

Activities: 1

System Administrator

Impact

3 - Low

Incident state

New

Opened by

System Administrator

Priority

4 - Low

Field changes • 2026-06-27 07:41:39

Update

Resolve

Delete

Responses

Related Links

Show SLA Timeline
Repair SLAs

Task SLAs (2)

Affected CIs

Impacted Services/CIs

Child Incidents

SLA definition

Search

Task = INC0010040

SLA definition

Type

Target

Stage

Business time left

Business elapsed time

Business elapsed percentage

Start time

Stop time

1 to 2 of 2

Figure: Incident Generated